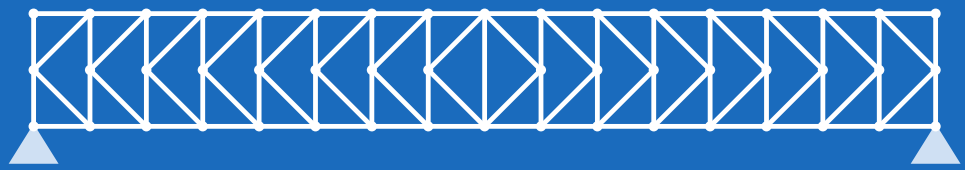
TRUSS REPORT

K-truss bridge (97 members)

Plane truss — analysis and CIRSOC 301 rod checks



Author A. Student · Course Structures I · 13/10/2026

MAX TENSION	MAX COMPRESSION	DEFLECTION	MOST USED ROD	BUCKLING Λ_1	WEIGHT
+945.0 kN	-945.0 kN	84.1 mm	0.73	2.34	15679 kg
rod 8	rod 24	L/761 · limit L/300	rod 8 · normal stress (H.3.4)	the load can grow ×2.3	carries 900.0 kN · 6× own weight

The model

Span	64.00 m
Height	8.00 m
Nodes / rods	50 / 97
Supports	pin at 0, rollerX at 16
Loads	15 nodal, total 900.0 kN
Rod families	5, with their sections:
Bottom chord	HEA 220, F24 (CIRSOC) / A36
Diagonal	SHS 150x150x6, F24 (CIRSOC) / A36
End post	SHS 200x200x8, F24 (CIRSOC) / A36
Top chord	HEB 220, F24 (CIRSOC) / A36
Vertical	SHS 120x120x5, F24 (CIRSOC) / A36

What the analysis found

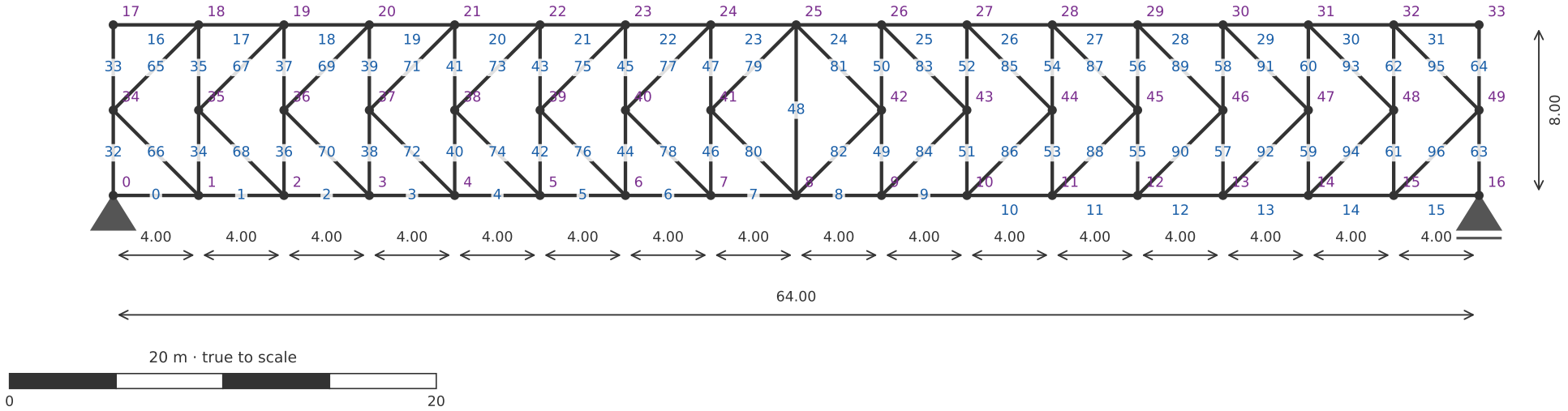
Stability	Statically determinate (m + r − 2j = 97 + 3 − 2×50 = 0): statics alone gives every force.
Most used rod	rod 8 at 0.73 (normal stress (H.3.4))
Rods over capacity	0 of 97 checked
Buckling	$\lambda_1 = 2.34$
Shear and moment	none: every rod is pinned and loaded only at its nodes, so the axial force is the whole result

Contents

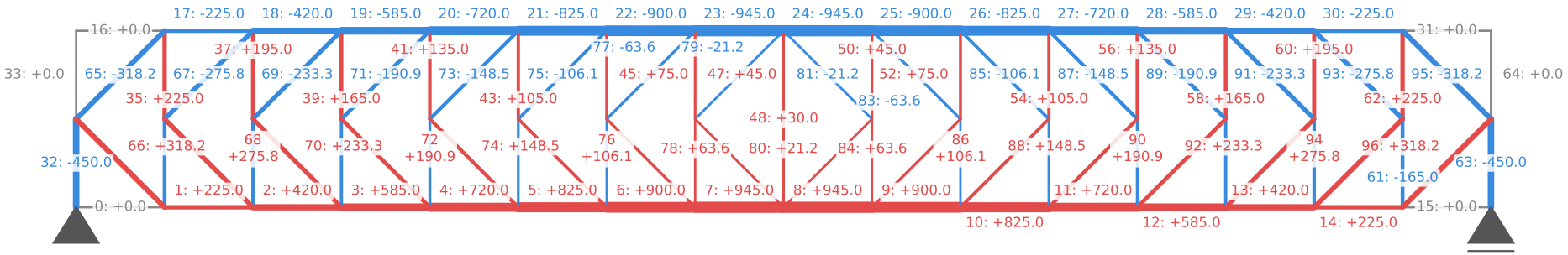
2 Geometry
3 Axial forces
4 Deflection and stress
5 Reactions
6 Utilisation
7 Buckling
8 Node force vectors · 7 sheets
15 Rods · 2 sheets
17 Rod calculations · 7 sheets
24 Explain rod 8 and 7

Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

Node numbers in purple, rod numbers in blue. Dimensions in m.



Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 945.0).

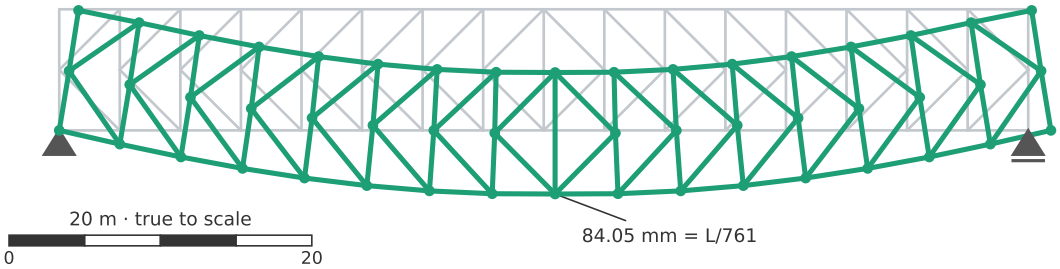


20 m · true to scale

0 20

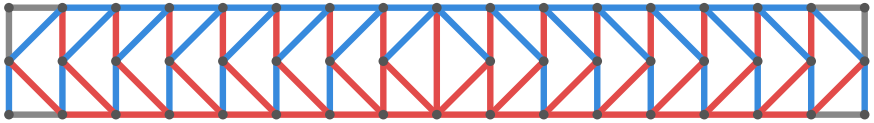
14 of 97 labels left out where they would overlap; every force is in the Rods table.

Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50. Every rod is pinned, so every member stays straight: there is no curvature, and no contraflexure. Largest vertical deflection 84.05 mm against the limit L/300 = 213.3 mm: OK.

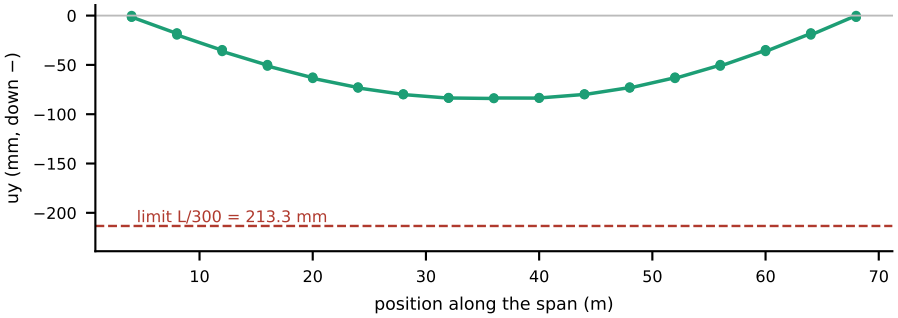


Fibre stress top and bottom face of each member

red = tension, blue = compression. Every rod is pinned, so both faces of a rod carry the same stress and each rod is one colour: the axial force.

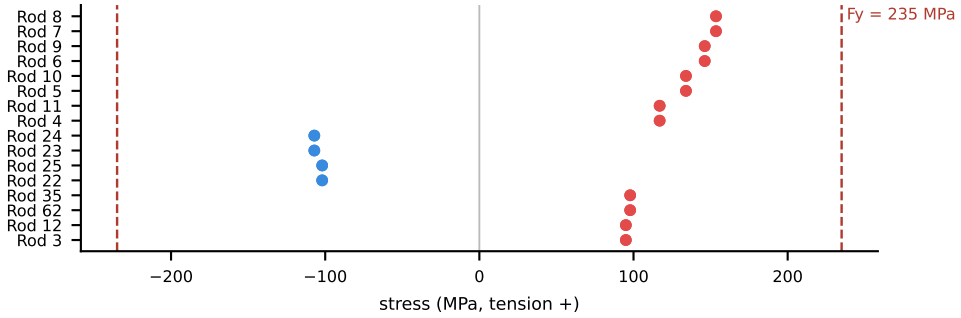


Deflection along the span vertical movement of every joint



Largest 84.05 mm at joint 8 = L/761. Within the limit.

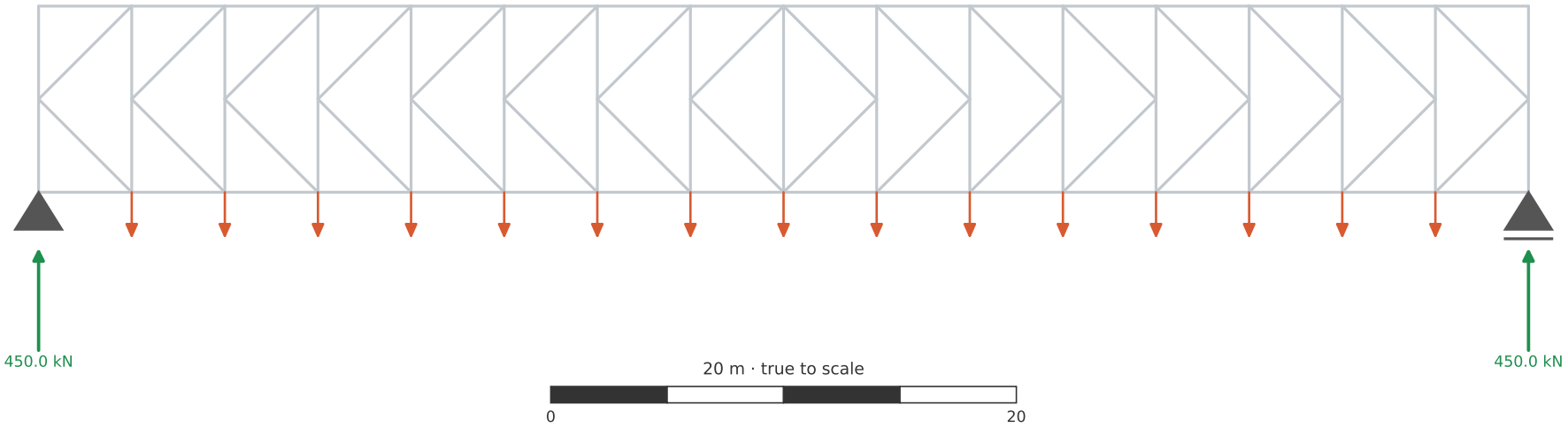
Fibre stress values the two faces of each rod, MPa



Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

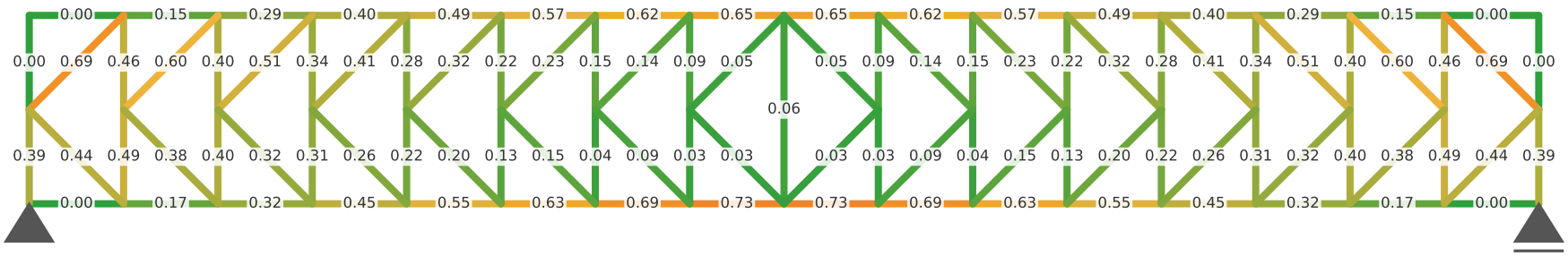
Support reactions (green arrows), as the supports push on the truss, against the loads (orange); Ry positive upwards. The sums balance the loads.

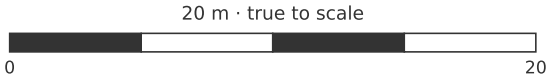
15 loads of 60.0 kN each, hung from the joints

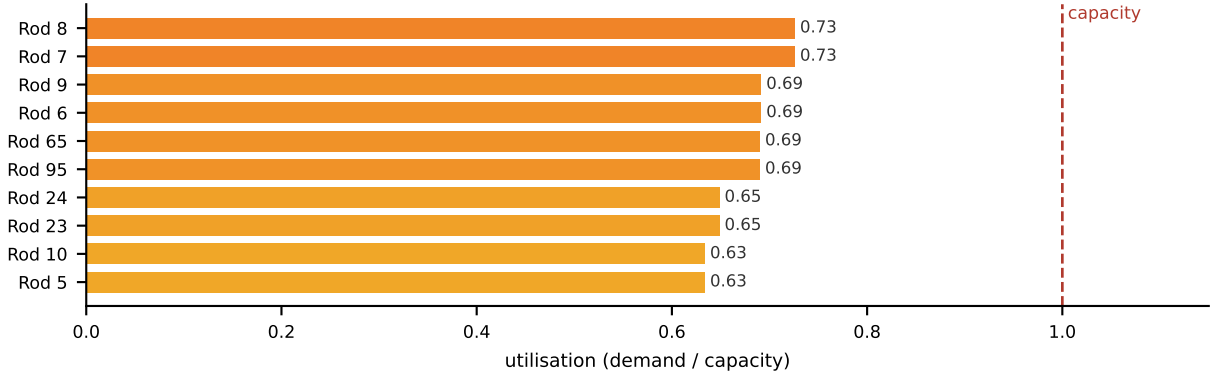


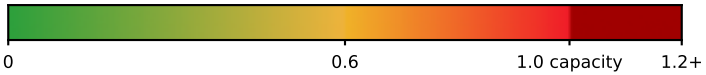
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	450.00
16	rollerX	0.00	450.00
Σ		0.00	900.00

Utilisation: demand over CIRSOC 301 capacity. Most used rod 8 at 0.73 (normal stress (H.3.4)). Every checked rod is within capacity.







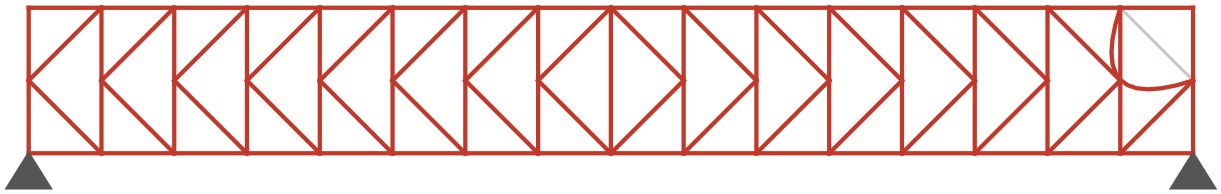


grey = no steel section, not checked

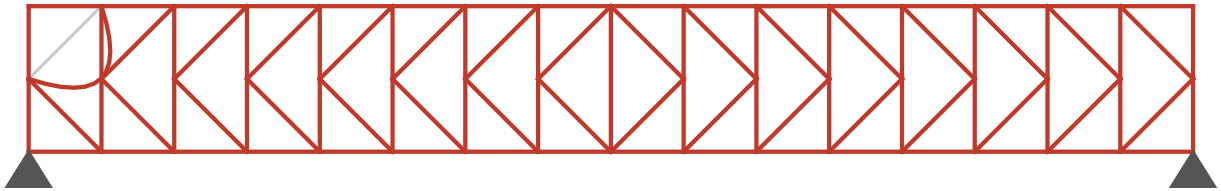
The 10 most used of 97 checked rods; the rest are in the Rods table.

λ is the factor on the load as it stands. $\lambda_1 = 2.34$, so the load can grow 2.3× before it buckles. 2 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

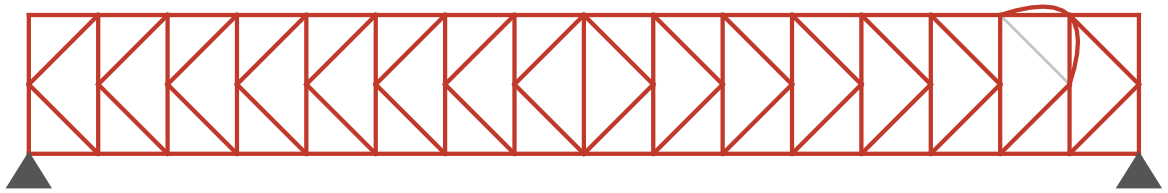
Mode 1: λ = 2.34
rod 95 bows

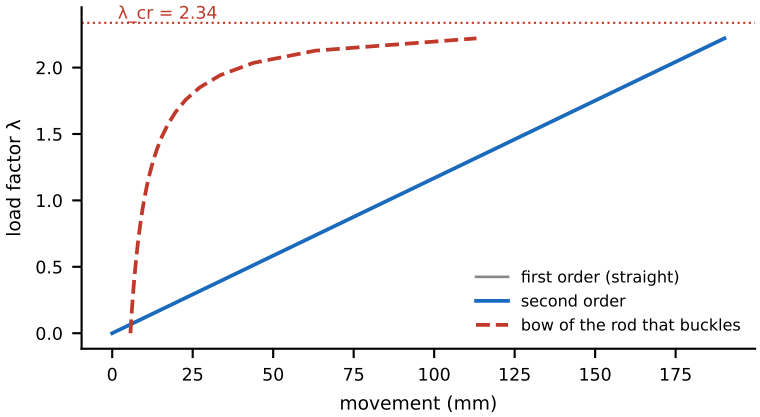


Mode 2: λ = 2.34
rod 65 bows



Mode 3: λ = 2.70
rod 93 bows





The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 5.66 mm in the first mode's shape.

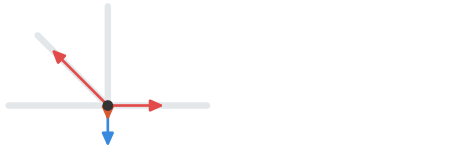
Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 0 · pin

member	N	θ°	Fx	Fy
r0→1	+0.0	180.0	+0.0	+0.0
r32→34	-450.0	270.0	+0.0	-450.0
reaction			+0.0	+450.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 1



member	N	θ°	Fx	Fy
r0→0	+0.0	0.0	+0.0	+0.0
r1→2	+225.0	0.0	+225.0	+0.0
r34→35	-165.0	270.0	+0.0	-165.0
r66→34	+318.2	135.0	-225.0	+225.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 2



member	N	θ°	Fx	Fy
r1→1	+225.0	180.0	-225.0	+0.0
r2→3	+420.0	0.0	+420.0	+0.0
r36→36	-135.0	270.0	+0.0	-135.0
r68→35	+275.8	135.0	-195.0	+195.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 3



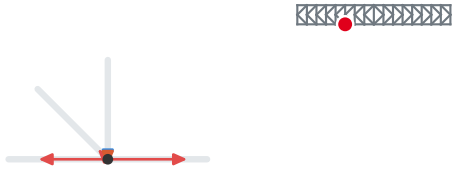
member	N	θ°	Fx	Fy
r2→2	+420.0	180.0	-420.0	+0.0
r3→4	+585.0	0.0	+585.0	+0.0
r38→37	-105.0	270.0	+0.0	-105.0
r70→36	+233.3	135.0	-165.0	+165.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 4



member	N	θ°	Fx	Fy
r3→3	+585.0	180.0	-585.0	+0.0
r4→5	+720.0	0.0	+720.0	+0.0
r40→38	-75.0	270.0	+0.0	-75.0
r72→37	+190.9	135.0	-135.0	+135.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 5



member	N	θ°	Fx	Fy
r4→4	+720.0	180.0	-720.0	+0.0
r5→6	+825.0	0.0	+825.0	+0.0
r42→39	-45.0	270.0	+0.0	-45.0
r74→38	+148.5	135.0	-105.0	+105.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 6



member	N	θ°	Fx	Fy
r5→5	+825.0	180.0	-825.0	+0.0
r6→7	+900.0	0.0	+900.0	+0.0
r44→40	-15.0	270.0	+0.0	-15.0
r76→39	+106.1	135.0	-75.0	+75.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 7



member	N	θ°	Fx	Fy
r6→6	+900.0	180.0	-900.0	+0.0
r7→8	+945.0	0.0	+945.0	+0.0
r46→41	+15.0	90.0	+0.0	+15.0
r78→40	+63.6	135.0	-45.0	+45.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

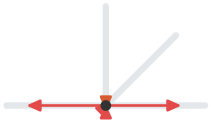
Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 8



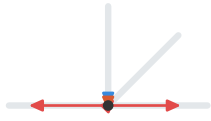
member	N	θ°	Fx	Fy
r7→7	+945.0	180.0	-945.0	+0.0
r8→9	+945.0	0.0	+945.0	+0.0
r48→25	+30.0	90.0	+0.0	+30.0
r80→41	+21.2	135.0	-15.0	+15.0
r82→42	+21.2	45.0	+15.0	+15.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 9



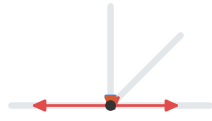
member	N	θ°	Fx	Fy
r8→8	+945.0	180.0	-945.0	+0.0
r9→10	+900.0	0.0	+900.0	+0.0
r49→42	+15.0	90.0	+0.0	+15.0
r84→43	+63.6	45.0	+45.0	+45.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 10



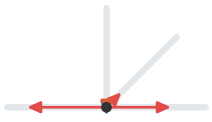
member	N	θ°	Fx	Fy
r9→9	+900.0	180.0	-900.0	+0.0
r10→11	+825.0	0.0	+825.0	+0.0
r51→43	-15.0	270.0	+0.0	-15.0
r86→44	+106.1	45.0	+75.0	+75.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 11



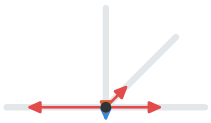
member	N	θ°	Fx	Fy
r10→10	+825.0	180.0	-825.0	+0.0
r11→12	+720.0	0.0	+720.0	+0.0
r53→44	-45.0	270.0	+0.0	-45.0
r88→45	+148.5	45.0	+105.0	+105.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 12



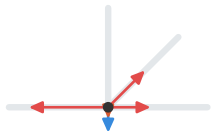
member	N	θ°	Fx	Fy
r11→11	+720.0	180.0	-720.0	+0.0
r12→13	+585.0	0.0	+585.0	+0.0
r55→45	-75.0	270.0	+0.0	-75.0
r90→46	+190.9	45.0	+135.0	+135.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 13



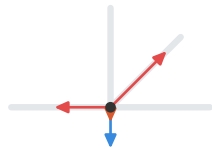
member	N	θ°	Fx	Fy
r12→12	+585.0	180.0	-585.0	+0.0
r13→14	+420.0	0.0	+420.0	+0.0
r57→46	-105.0	270.0	+0.0	-105.0
r92→47	+233.3	45.0	+165.0	+165.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 14



member	N	θ°	Fx	Fy
r13→13	+420.0	180.0	-420.0	+0.0
r14→15	+225.0	0.0	+225.0	+0.0
r59→47	-135.0	270.0	+0.0	-135.0
r94→48	+275.8	45.0	+195.0	+195.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 15



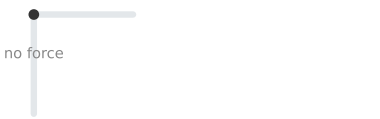
member	N	θ°	Fx	Fy
r14→14	+225.0	180.0	-225.0	+0.0
r15→16	+0.0	180.0	+0.0	+0.0
r61→48	-165.0	270.0	+0.0	-165.0
r96→49	+318.2	45.0	+225.0	+225.0
load			+0.0	-60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 16 · rollerX

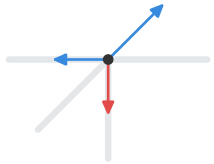
member	N	θ°	Fx	Fy
r15→15	+0.0	0.0	+0.0	+0.0
r63→49	-450.0	270.0	+0.0	-450.0
reaction			+0.0	+450.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 17



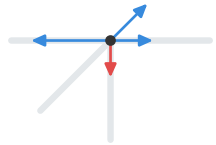
member	N	θ°	Fx	Fy
r16→18	+0.0	0.0	+0.0	+0.0
r33→34	+0.0	180.0	+0.0	+0.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 18



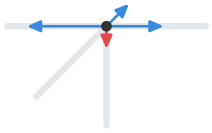
member	N	θ°	Fx	Fy
r16→17	+0.0	180.0	+0.0	+0.0
r17→19	-225.0	180.0	-225.0	+0.0
r35→35	+225.0	270.0	+0.0	-225.0
r65→34	-318.2	45.0	+225.0	+225.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 19



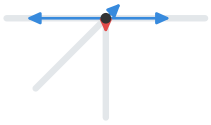
member	N	θ°	Fx	Fy
r17→18	-225.0	0.0	+225.0	+0.0
r18→20	-420.0	180.0	-420.0	+0.0
r37→36	+195.0	270.0	+0.0	-195.0
r67→35	-275.8	45.0	+195.0	+195.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 20



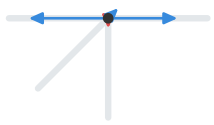
member	N	θ°	Fx	Fy
r18→19	-420.0	0.0	+420.0	+0.0
r19→21	-585.0	180.0	-585.0	+0.0
r39→37	+165.0	270.0	+0.0	-165.0
r69→36	-233.3	45.0	+165.0	+165.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 21



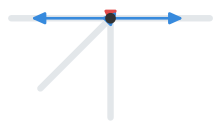
member	N	θ°	Fx	Fy
r19→20	-585.0	0.0	+585.0	+0.0
r20→22	-720.0	180.0	-720.0	+0.0
r41→38	+135.0	270.0	+0.0	-135.0
r71→37	-190.9	45.0	+135.0	+135.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 22



member	N	θ°	Fx	Fy
r20→21	-720.0	0.0	+720.0	+0.0
r21→23	-825.0	180.0	-825.0	+0.0
r43→39	+105.0	270.0	+0.0	-105.0
r73→38	-148.5	45.0	+105.0	+105.0
Σ kN	Fx 0.00	Fy 0.00	✓	

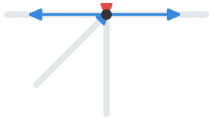
Joint 23



member	N	θ°	Fx	Fy
r21→22	-825.0	0.0	+825.0	+0.0
r22→24	-900.0	180.0	-900.0	+0.0
r45→40	+75.0	270.0	+0.0	-75.0
r75→39	-106.1	45.0	+75.0	+75.0
Σ kN	Fx 0.00	Fy 0.00	✓	

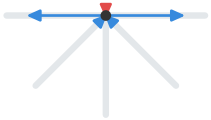
Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 24



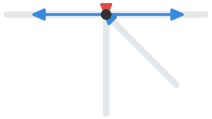
member	N	θ°	Fx	Fy
r22→23	-900.0	0.0	+900.0	+0.0
r23→25	-945.0	180.0	-945.0	+0.0
r47→41	+45.0	270.0	+0.0	-45.0
r77→40	-63.6	45.0	+45.0	+45.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 25



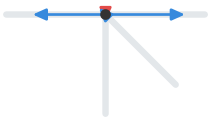
member	N	θ°	Fx	Fy
r23→24	-945.0	0.0	+945.0	+0.0
r24→26	-945.0	180.0	-945.0	+0.0
r48→8	+30.0	270.0	+0.0	-30.0
r79→41	-21.2	45.0	+15.0	+15.0
r81→42	-21.2	135.0	-15.0	+15.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 26



member	N	θ°	Fx	Fy
r24→25	-945.0	0.0	+945.0	+0.0
r25→27	-900.0	180.0	-900.0	+0.0
r50→42	+45.0	270.0	+0.0	-45.0
r83→43	-63.6	135.0	-45.0	+45.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 27



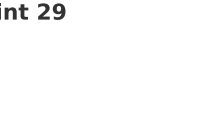
member	N	θ°	Fx	Fy
r25→26	-900.0	0.0	+900.0	+0.0
r26→28	-825.0	180.0	-825.0	+0.0
r52→43	+75.0	270.0	+0.0	-75.0
r85→44	-106.1	135.0	-75.0	+75.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 28



member	N	θ°	Fx	Fy
r26→27	-825.0	0.0	+825.0	+0.0
r27→29	-720.0	180.0	-720.0	+0.0
r54→44	+105.0	270.0	+0.0	-105.0
r87→45	-148.5	135.0	-105.0	+105.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 29



member	N	θ°	Fx	Fy
r27→28	-720.0	0.0	+720.0	+0.0
r28→30	-585.0	180.0	-585.0	+0.0
r56→45	+135.0	270.0	+0.0	-135.0
r89→46	-190.9	135.0	-135.0	+135.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 30



member	N	θ°	Fx	Fy
r28→29	-585.0	0.0	+585.0	+0.0
r29→31	-420.0	180.0	-420.0	+0.0
r58→46	+165.0	270.0	+0.0	-165.0
r91→47	-233.3	135.0	-165.0	+165.0
Σ kN	Fx 0.00	Fy 0.00	✓	

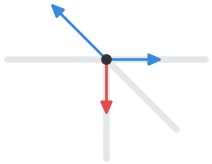
Joint 31



member	N	θ°	Fx	Fy
r29→30	-420.0	0.0	+420.0	+0.0
r30→32	-225.0	180.0	-225.0	+0.0
r60→47	+195.0	270.0	+0.0	-195.0
r93→48	-275.8	135.0	-195.0	+195.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 32



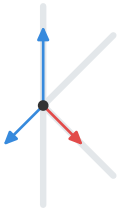
member	N	θ°	Fx	Fy
r30→31	-225.0	0.0	+225.0	+0.0
r31→33	+0.0	180.0	+0.0	+0.0
r62→48	+225.0	270.0	+0.0	-225.0
r95→49	-318.2	135.0	-225.0	+225.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 33



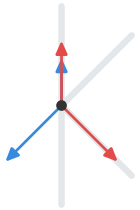
member	N	θ°	Fx	Fy
r31→32	+0.0	0.0	+0.0	+0.0
r64→49	+0.0	180.0	+0.0	+0.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 34



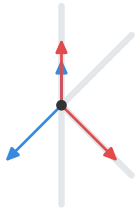
member	N	θ°	Fx	Fy
r32→0	-450.0	90.0	+0.0	+450.0
r33→17	+0.0	180.0	+0.0	+0.0
r65→18	-318.2	225.0	-225.0	-225.0
r66→1	+318.2	315.0	+225.0	-225.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 35



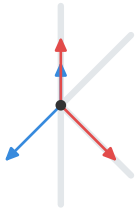
member	N	θ°	Fx	Fy
r34→1	-165.0	90.0	+0.0	+165.0
r35→18	+225.0	90.0	+0.0	+225.0
r67→19	-275.8	225.0	-195.0	-195.0
r68→2	+275.8	315.0	+195.0	-195.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 36



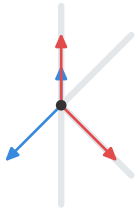
member	N	θ°	Fx	Fy
r36→2	-135.0	90.0	+0.0	+135.0
r37→19	+195.0	90.0	+0.0	+195.0
r69→20	-233.3	225.0	-165.0	-165.0
r70→3	+233.3	315.0	+165.0	-165.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 37



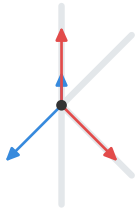
member	N	θ°	Fx	Fy
r38→3	-105.0	90.0	+0.0	+105.0
r39→20	+165.0	90.0	+0.0	+165.0
r71→21	-190.9	225.0	-135.0	-135.0
r72→4	+190.9	315.0	+135.0	-135.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 38



member	N	θ°	Fx	Fy
r40→4	-75.0	90.0	+0.0	+75.0
r41→21	+135.0	90.0	+0.0	+135.0
r73→22	-148.5	225.0	-105.0	-105.0
r74→5	+148.5	315.0	+105.0	-105.0
Σ kN	Fx 0.00	Fy 0.00	✓	

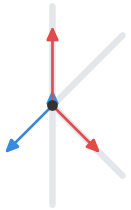
Joint 39



member	N	θ°	Fx	Fy
r42→5	-45.0	90.0	+0.0	+45.0
r43→22	+105.0	90.0	+0.0	+105.0
r75→23	-106.1	225.0	-75.0	-75.0
r76→6	+106.1	315.0	+75.0	-75.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 40



member	N	θ°	Fx	Fy
r44→6	-15.0	90.0	+0.0	+15.0
r45→23	+75.0	90.0	+0.0	+75.0
r77→24	-63.6	225.0	-45.0	-45.0
r78→7	+63.6	315.0	+45.0	-45.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 41



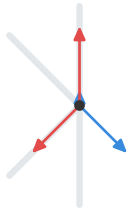
member	N	θ°	Fx	Fy
r46→7	+15.0	270.0	+0.0	-15.0
r47→24	+45.0	90.0	+0.0	+45.0
r79→25	-21.2	225.0	-15.0	-15.0
r80→8	+21.2	315.0	+15.0	-15.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 42



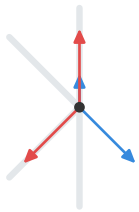
member	N	θ°	Fx	Fy
r49→9	+15.0	270.0	+0.0	-15.0
r50→26	+45.0	90.0	+0.0	+45.0
r81→25	-21.2	315.0	+15.0	-15.0
r82→8	+21.2	225.0	-15.0	-15.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 43



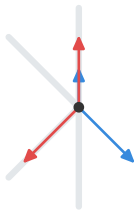
member	N	θ°	Fx	Fy
r51→10	-15.0	90.0	+0.0	+15.0
r52→27	+75.0	90.0	+0.0	+75.0
r83→26	-63.6	315.0	+45.0	-45.0
r84→9	+63.6	225.0	-45.0	-45.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 44



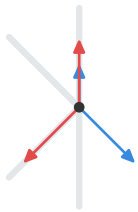
member	N	θ°	Fx	Fy
r53→11	-45.0	90.0	+0.0	+45.0
r54→28	+105.0	90.0	+0.0	+105.0
r85→27	-106.1	315.0	+75.0	-75.0
r86→10	+106.1	225.0	-75.0	-75.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 45



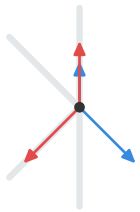
member	N	θ°	Fx	Fy
r55→12	-75.0	90.0	+0.0	+75.0
r56→29	+135.0	90.0	+0.0	+135.0
r87→28	-148.5	315.0	+105.0	-105.0
r88→11	+148.5	225.0	-105.0	-105.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 46



member	N	θ°	Fx	Fy
r57→13	-105.0	90.0	+0.0	+105.0
r58→30	+165.0	90.0	+0.0	+165.0
r89→29	-190.9	315.0	+135.0	-135.0
r90→12	+190.9	225.0	-135.0	-135.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 47



member	N	θ°	Fx	Fy
r59→14	-135.0	90.0	+0.0	+135.0
r60→31	+195.0	90.0	+0.0	+195.0
r91→30	-233.3	315.0	+165.0	-165.0
r92→13	+233.3	225.0	-165.0	-165.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 48



member	N	θ°	Fx	Fy
r61→15	-165.0	90.0	+0.0	+165.0
r62→32	+225.0	90.0	+0.0	+225.0
r93→31	-275.8	315.0	+195.0	-195.0
r94→14	+275.8	225.0	-195.0	-195.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 49



member	N	θ°	Fx	Fy
r63→16	-450.0	90.0	+0.0	+450.0
r64→33	+0.0	180.0	+0.0	+0.0
r95→32	-318.2	315.0	+225.0	-225.0
r96→15	+318.2	225.0	-225.0	-225.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
0	0–1	Bottom chord	HEA 220	4.00	+0.00	inelastic buckling (E3-2)	0.00 OK
1	1–2	Bottom chord	HEA 220	4.00	+225.00	normal stress (H.3.4)	0.17 OK
2	2–3	Bottom chord	HEA 220	4.00	+420.00	normal stress (H.3.4)	0.32 OK
3	3–4	Bottom chord	HEA 220	4.00	+585.00	normal stress (H.3.4)	0.45 OK
4	4–5	Bottom chord	HEA 220	4.00	+720.00	normal stress (H.3.4)	0.55 OK
5	5–6	Bottom chord	HEA 220	4.00	+825.00	normal stress (H.3.4)	0.63 OK
6	6–7	Bottom chord	HEA 220	4.00	+900.00	normal stress (H.3.4)	0.69 OK
7	7–8	Bottom chord	HEA 220	4.00	+945.00	normal stress (H.3.4)	0.73 OK
8	8–9	Bottom chord	HEA 220	4.00	+945.00	normal stress (H.3.4)	0.73 OK
9	9–10	Bottom chord	HEA 220	4.00	+900.00	normal stress (H.3.4)	0.69 OK
10	10–11	Bottom chord	HEA 220	4.00	+825.00	normal stress (H.3.4)	0.63 OK
11	11–12	Bottom chord	HEA 220	4.00	+720.00	normal stress (H.3.4)	0.55 OK
12	12–13	Bottom chord	HEA 220	4.00	+585.00	normal stress (H.3.4)	0.45 OK
13	13–14	Bottom chord	HEA 220	4.00	+420.00	normal stress (H.3.4)	0.32 OK
14	14–15	Bottom chord	HEA 220	4.00	+225.00	normal stress (H.3.4)	0.17 OK
15	15–16	Bottom chord	HEA 220	4.00	+0.00	inelastic buckling (E3-2)	0.00 OK
16	17–18	Top chord	HEB 220	4.00	+0.00	normal stress (H.3.4)	0.00 OK
17	18–19	Top chord	HEB 220	4.00	-225.00	inelastic buckling (E3-2)	0.15 OK
18	19–20	Top chord	HEB 220	4.00	-420.00	inelastic buckling (E3-2)	0.29 OK
19	20–21	Top chord	HEB 220	4.00	-585.00	inelastic buckling (E3-2)	0.40 OK
20	21–22	Top chord	HEB 220	4.00	-720.00	inelastic buckling (E3-2)	0.49 OK
21	22–23	Top chord	HEB 220	4.00	-825.00	inelastic buckling (E3-2)	0.57 OK
22	23–24	Top chord	HEB 220	4.00	-900.00	inelastic buckling (E3-2)	0.62 OK
23	24–25	Top chord	HEB 220	4.00	-945.00	inelastic buckling (E3-2)	0.65 OK
24	25–26	Top chord	HEB 220	4.00	-945.00	inelastic buckling (E3-2)	0.65 OK
25	26–27	Top chord	HEB 220	4.00	-900.00	inelastic buckling (E3-2)	0.62 OK
26	27–28	Top chord	HEB 220	4.00	-825.00	inelastic buckling (E3-2)	0.57 OK
27	28–29	Top chord	HEB 220	4.00	-720.00	inelastic buckling (E3-2)	0.49 OK
28	29–30	Top chord	HEB 220	4.00	-585.00	inelastic buckling (E3-2)	0.40 OK
29	30–31	Top chord	HEB 220	4.00	-420.00	inelastic buckling (E3-2)	0.29 OK
30	31–32	Top chord	HEB 220	4.00	-225.00	inelastic buckling (E3-2)	0.15 OK
31	32–33	Top chord	HEB 220	4.00	+0.00	inelastic buckling (E3-2)	0.00 OK
32	0–34	End post	SHS 200x200x8	4.00	-450.00	inelastic buckling (E3-2)	0.39 OK
33	34–17	End post	SHS 200x200x8	4.00	+0.00	normal stress (H.3.4)	0.00 OK
34	1–35	Vertical	SHS 120x120x5	4.00	-165.00	inelastic buckling (E3-2)	0.49 OK
35	35–18	Vertical	SHS 120x120x5	4.00	+225.00	normal stress (H.3.4)	0.46 OK
36	2–36	Vertical	SHS 120x120x5	4.00	-135.00	inelastic buckling (E3-2)	0.40 OK
37	36–19	Vertical	SHS 120x120x5	4.00	+195.00	normal stress (H.3.4)	0.40 OK
38	3–37	Vertical	SHS 120x120x5	4.00	-105.00	inelastic buckling (E3-2)	0.31 OK
39	37–20	Vertical	SHS 120x120x5	4.00	+165.00	normal stress (H.3.4)	0.34 OK
40	4–38	Vertical	SHS 120x120x5	4.00	-75.00	inelastic buckling (E3-2)	0.22 OK
41	38–21	Vertical	SHS 120x120x5	4.00	+135.00	normal stress (H.3.4)	0.28 OK
42	5–39	Vertical	SHS 120x120x5	4.00	-45.00	inelastic buckling (E3-2)	0.13 OK
43	39–22	Vertical	SHS 120x120x5	4.00	+105.00	normal stress (H.3.4)	0.22 OK
44	6–40	Vertical	SHS 120x120x5	4.00	-15.00	inelastic buckling (E3-2)	0.04 OK
45	40–23	Vertical	SHS 120x120x5	4.00	+75.00	normal stress (H.3.4)	0.15 OK
46	7–41	Vertical	SHS 120x120x5	4.00	+15.00	normal stress (H.3.4)	0.03 OK
47	41–24	Vertical	SHS 120x120x5	4.00	+45.00	normal stress (H.3.4)	0.09 OK
48	8–25	Vertical	SHS 120x120x5	8.00	+30.00	normal stress (H.3.4)	0.06 OK
49	9–42	Vertical	SHS 120x120x5	4.00	+15.00	normal stress (H.3.4)	0.03 OK
50	42–26	Vertical	SHS 120x120x5	4.00	+45.00	normal stress (H.3.4)	0.09 OK
51	10–43	Vertical	SHS 120x120x5	4.00	-15.00	inelastic buckling (E3-2)	0.04 OK

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
52	43–27	Vertical	SHS 120x120x5	4.00	+75.00	normal stress (H.3.4)	0.15 OK
53	11–44	Vertical	SHS 120x120x5	4.00	-45.00	inelastic buckling (E3-2)	0.13 OK
54	44–28	Vertical	SHS 120x120x5	4.00	+105.00	normal stress (H.3.4)	0.22 OK
55	12–45	Vertical	SHS 120x120x5	4.00	-75.00	inelastic buckling (E3-2)	0.22 OK
56	45–29	Vertical	SHS 120x120x5	4.00	+135.00	normal stress (H.3.4)	0.28 OK
57	13–46	Vertical	SHS 120x120x5	4.00	-105.00	inelastic buckling (E3-2)	0.31 OK
58	46–30	Vertical	SHS 120x120x5	4.00	+165.00	normal stress (H.3.4)	0.34 OK
59	14–47	Vertical	SHS 120x120x5	4.00	-135.00	inelastic buckling (E3-2)	0.40 OK
60	47–31	Vertical	SHS 120x120x5	4.00	+195.00	normal stress (H.3.4)	0.40 OK
61	15–48	Vertical	SHS 120x120x5	4.00	-165.00	inelastic buckling (E3-2)	0.49 OK
62	48–32	Vertical	SHS 120x120x5	4.00	+225.00	normal stress (H.3.4)	0.46 OK
63	16–49	End post	SHS 200x200x8	4.00	-450.00	inelastic buckling (E3-2)	0.39 OK
64	49–33	End post	SHS 200x200x8	4.00	+0.00	normal stress (H.3.4)	0.00 OK
65	34–18	Diagonal	SHS 150x150x6	5.66	-318.20	inelastic buckling (E3-2)	0.69 OK
66	34–1	Diagonal	SHS 150x150x6	5.66	+318.20	normal stress (H.3.4)	0.44 OK
67	35–19	Diagonal	SHS 150x150x6	5.66	-275.77	inelastic buckling (E3-2)	0.60 OK
68	35–2	Diagonal	SHS 150x150x6	5.66	+275.77	normal stress (H.3.4)	0.38 OK
69	36–20	Diagonal	SHS 150x150x6	5.66	-233.35	inelastic buckling (E3-2)	0.51 OK
70	36–3	Diagonal	SHS 150x150x6	5.66	+233.35	normal stress (H.3.4)	0.32 OK
71	37–21	Diagonal	SHS 150x150x6	5.66	-190.92	inelastic buckling (E3-2)	0.41 OK
72	37–4	Diagonal	SHS 150x150x6	5.66	+190.92	normal stress (H.3.4)	0.26 OK
73	38–22	Diagonal	SHS 150x150x6	5.66	-148.49	inelastic buckling (E3-2)	0.32 OK
74	38–5	Diagonal	SHS 150x150x6	5.66	+148.49	normal stress (H.3.4)	0.20 OK
75	39–23	Diagonal	SHS 150x150x6	5.66	-106.07	inelastic buckling (E3-2)	0.23 OK
76	39–6	Diagonal	SHS 150x150x6	5.66	+106.07	normal stress (H.3.4)	0.15 OK
77	40–24	Diagonal	SHS 150x150x6	5.66	-63.64	inelastic buckling (E3-2)	0.14 OK
78	40–7	Diagonal	SHS 150x150x6	5.66	+63.64	normal stress (H.3.4)	0.09 OK
79	41–25	Diagonal	SHS 150x150x6	5.66	-21.21	inelastic buckling (E3-2)	0.05 OK
80	41–8	Diagonal	SHS 150x150x6	5.66	+21.21	normal stress (H.3.4)	0.03 OK
81	42–25	Diagonal	SHS 150x150x6	5.66	-21.21	inelastic buckling (E3-2)	0.05 OK
82	42–8	Diagonal	SHS 150x150x6	5.66	+21.21	normal stress (H.3.4)	0.03 OK
83	43–26	Diagonal	SHS 150x150x6	5.66	-63.64	inelastic buckling (E3-2)	0.14 OK
84	43–9	Diagonal	SHS 150x150x6	5.66	+63.64	normal stress (H.3.4)	0.09 OK
85	44–27	Diagonal	SHS 150x150x6	5.66	-106.07	inelastic buckling (E3-2)	0.23 OK
86	44–10	Diagonal	SHS 150x150x6	5.66	+106.07	normal stress (H.3.4)	0.15 OK
87	45–28	Diagonal	SHS 150x150x6	5.66	-148.49	inelastic buckling (E3-2)	0.32 OK
88	45–11	Diagonal	SHS 150x150x6	5.66	+148.49	normal stress (H.3.4)	0.20 OK
89	46–29	Diagonal	SHS 150x150x6	5.66	-190.92	inelastic buckling (E3-2)	0.41 OK
90	46–12	Diagonal	SHS 150x150x6	5.66	+190.92	normal stress (H.3.4)	0.26 OK
91	47–30	Diagonal	SHS 150x150x6	5.66	-233.35	inelastic buckling (E3-2)	0.51 OK
92	47–13	Diagonal	SHS 150x150x6	5.66	+233.35	normal stress (H.3.4)	0.32 OK
93	48–31	Diagonal	SHS 150x150x6	5.66	-275.77	inelastic buckling (E3-2)	0.60 OK
94	48–14	Diagonal	SHS 150x150x6	5.66	+275.77	normal stress (H.3.4)	0.38 OK
95	49–32	Diagonal	SHS 150x150x6	5.66	-318.20	inelastic buckling (E3-2)	0.69 OK
96	49–15	Diagonal	SHS 150x150x6	5.66	+318.20	normal stress (H.3.4)	0.44 OK

Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 0

joint 0 → 1

no force



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +0.00 kN
at 0:  Fx = +0.00   Fy = +0.00
at 1:  Fx = +0.00   Fy = +0.00
```



Rod 1

joint 1 → 2

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +225.00 kN
at 1:  Fx = +225.00   Fy = +0.00
at 2:  Fx = -225.00   Fy = +0.00
```



Rod 2

joint 2 → 3

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +420.00 kN
at 2:  Fx = +420.00   Fy = +0.00
at 3:  Fx = -420.00   Fy = +0.00
```



Rod 3

joint 3 → 4

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +585.00 kN
at 3:  Fx = +585.00   Fy = +0.00
at 4:  Fx = -585.00   Fy = +0.00
```



Rod 4

joint 4 → 5

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +720.00 kN
at 4:  Fx = +720.00   Fy = +0.00
at 5:  Fx = -720.00   Fy = +0.00
```



Rod 5

joint 5 → 6

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +825.00 kN
at 5:  Fx = +825.00   Fy = +0.00
at 6:  Fx = -825.00   Fy = +0.00
```



Rod 6

joint 6 → 7

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +900.00 kN
at 6:  Fx = +900.00   Fy = +0.00
at 7:  Fx = -900.00   Fy = +0.00
```



Rod 7

joint 7 → 8

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +945.00 kN
at 7:  Fx = +945.00   Fy = +0.00
at 8:  Fx = -945.00   Fy = +0.00
```



Rod 8

joint 8 → 9

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +945.00 kN
at 8:  Fx = +945.00   Fy = +0.00
at 9:  Fx = -945.00   Fy = +0.00
```



Rod 9

joint 9 → 10

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +900.00 kN
at 9:  Fx = +900.00   Fy = +0.00
at 10:  Fx = -900.00   Fy = +0.00
```



Rod 10

joint 10 → 11

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +825.00 kN
at 10:  Fx = +825.00   Fy = +0.00
at 11:  Fx = -825.00   Fy = +0.00
```



Rod 11

joint 11 → 12

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +720.00 kN
at 11:  Fx = +720.00   Fy = +0.00
at 12:  Fx = -720.00   Fy = +0.00
```



Rod 12

joint 12 → 13

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +585.00 kN
at 12:  Fx = +585.00   Fy = +0.00
at 13:  Fx = -585.00   Fy = +0.00
```



Rod 13

joint 13 → 14

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +420.00 kN
at 13:  Fx = +420.00   Fy = +0.00
at 14:  Fx = -420.00   Fy = +0.00
```



Rod 14

joint 14 → 15

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +225.00 kN
at 14:  Fx = +225.00   Fy = +0.00
at 15:  Fx = -225.00   Fy = +0.00
```



Rod 15

joint 15 → 16

no force



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +0.00 kN
at 15:  Fx = +0.00   Fy = +0.00
at 16:  Fx = +0.00   Fy = +0.00
```



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

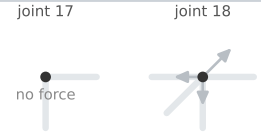
Rod 16

joint 17 → 18

no force



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = +0.00 kN
at 17: Fx = +0.00 Fy = +0.00
at 18: Fx = +0.00 Fy = +0.00



Rod 17

joint 18 → 19

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -225.00 kN
at 18: Fx = -225.00 Fy = +0.00
at 19: Fx = +225.00 Fy = +0.00



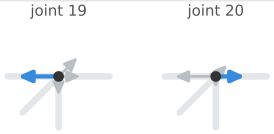
Rod 18

joint 19 → 20

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -420.00 kN
at 19: Fx = -420.00 Fy = +0.00
at 20: Fx = +420.00 Fy = +0.00



Rod 19

joint 20 → 21

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -585.00 kN
at 20: Fx = -585.00 Fy = +0.00
at 21: Fx = +585.00 Fy = +0.00



Rod 20

joint 21 → 22

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -720.00 kN
at 21: Fx = -720.00 Fy = +0.00
at 22: Fx = +720.00 Fy = +0.00



Rod 21

joint 22 → 23

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -825.00 kN
at 22: Fx = -825.00 Fy = +0.00
at 23: Fx = +825.00 Fy = +0.00



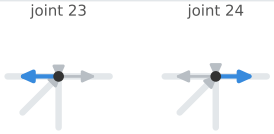
Rod 22

joint 23 → 24

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -900.00 kN
at 23: Fx = -900.00 Fy = +0.00
at 24: Fx = +900.00 Fy = +0.00



Rod 23

joint 24 → 25

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -945.00 kN
at 24: Fx = -945.00 Fy = +0.00
at 25: Fx = +945.00 Fy = +0.00



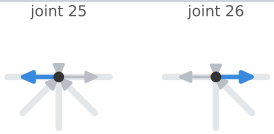
Rod 24

joint 25 → 26

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -945.00 kN
at 25: Fx = -945.00 Fy = +0.00
at 26: Fx = +945.00 Fy = +0.00



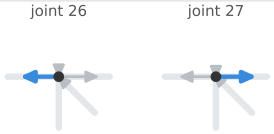
Rod 25

joint 26 → 27

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -900.00 kN
at 26: Fx = -900.00 Fy = +0.00
at 27: Fx = +900.00 Fy = +0.00



Rod 26

joint 27 → 28

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -825.00 kN
at 27: Fx = -825.00 Fy = +0.00
at 28: Fx = +825.00 Fy = +0.00



Rod 27

joint 28 → 29

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -720.00 kN
at 28: Fx = -720.00 Fy = +0.00
at 29: Fx = +720.00 Fy = +0.00



Rod 28

joint 29 → 30

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -585.00 kN
at 29: Fx = -585.00 Fy = +0.00
at 30: Fx = +585.00 Fy = +0.00



Rod 29

joint 30 → 31

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -420.00 kN
at 30: Fx = -420.00 Fy = +0.00
at 31: Fx = +420.00 Fy = +0.00



Rod 30

joint 31 → 32

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -225.00 kN
at 31: Fx = -225.00 Fy = +0.00
at 32: Fx = +225.00 Fy = +0.00



Rod 31

joint 32 → 33

no force



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = +0.00 kN
at 32: Fx = +0.00 Fy = +0.00
at 33: Fx = +0.00 Fy = +0.00



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 32

joint 0 → 34

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -450.00 kN
at 0: Fx = +0.00 Fy = -450.00
at 34: Fx = +0.00 Fy = +450.00

joint 0

joint 34



Rod 33

joint 34 → 17

no force



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +0.00 kN
at 34: Fx = +0.00 Fy = +0.00
at 17: Fx = +0.00 Fy = +0.00

joint 34



joint 17



Rod 34

joint 1 → 35

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -165.00 kN
at 1: Fx = +0.00 Fy = -165.00
at 35: Fx = +0.00 Fy = +165.00

joint 1



joint 35



Rod 35

joint 35 → 18

tension

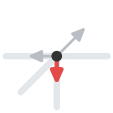


L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +225.00 kN
at 35: Fx = +0.00 Fy = +225.00
at 18: Fx = +0.00 Fy = -225.00

joint 35



joint 18



Rod 36

joint 2 → 36

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -135.00 kN
at 2: Fx = +0.00 Fy = -135.00
at 36: Fx = +0.00 Fy = +135.00

joint 2



joint 36



Rod 37

joint 36 → 19

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +195.00 kN
at 36: Fx = +0.00 Fy = +195.00
at 19: Fx = +0.00 Fy = -195.00

joint 36



joint 19



Rod 38

joint 3 → 37

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -105.00 kN
at 3: Fx = +0.00 Fy = -105.00
at 37: Fx = +0.00 Fy = +105.00

joint 3



joint 37



Rod 39

joint 37 → 20

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +165.00 kN
at 37: Fx = +0.00 Fy = +165.00
at 20: Fx = +0.00 Fy = -165.00

joint 37



joint 20



Rod 40

joint 4 → 38

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -75.00 kN
at 4: Fx = +0.00 Fy = -75.00
at 38: Fx = +0.00 Fy = +75.00

joint 4



joint 38



Rod 41

joint 38 → 21

tension

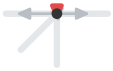


L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +135.00 kN
at 38: Fx = +0.00 Fy = +135.00
at 21: Fx = +0.00 Fy = -135.00

joint 38



joint 21



Rod 42

joint 5 → 39

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -45.00 kN
at 5: Fx = +0.00 Fy = -45.00
at 39: Fx = +0.00 Fy = +45.00

joint 5



joint 39



Rod 43

joint 39 → 22

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +105.00 kN
at 39: Fx = +0.00 Fy = +105.00
at 22: Fx = +0.00 Fy = -105.00

joint 39



joint 22



Rod 44

joint 6 → 40

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -15.00 kN
at 6: Fx = +0.00 Fy = -15.00
at 40: Fx = +0.00 Fy = +15.00

joint 6



joint 40



Rod 45

joint 40 → 23

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +75.00 kN
at 40: Fx = +0.00 Fy = +75.00
at 23: Fx = +0.00 Fy = -75.00

joint 40



joint 23



Rod 46

joint 7 → 41

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +15.00 kN
at 7: Fx = +0.00 Fy = +15.00
at 41: Fx = +0.00 Fy = -15.00

joint 7



joint 41



Rod 47

joint 41 → 24

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +45.00 kN
at 41: Fx = +0.00 Fy = +45.00
at 24: Fx = +0.00 Fy = -45.00

joint 41



joint 24



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 48

joint 8 → 25

tension



L = 8.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +30.00 kN
at 8: Fx = +0.00 Fy = +30.00
at 25: Fx = +0.00 Fy = -30.00

joint 8



joint 25



Rod 49

joint 9 → 42

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +15.00 kN
at 9: Fx = +0.00 Fy = +15.00
at 42: Fx = +0.00 Fy = -15.00

joint 9



joint 42



Rod 50

joint 42 → 26

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +45.00 kN
at 42: Fx = +0.00 Fy = +45.00
at 26: Fx = +0.00 Fy = -45.00

joint 42



joint 26



Rod 51

joint 10 → 43

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -15.00 kN
at 10: Fx = +0.00 Fy = -15.00
at 43: Fx = +0.00 Fy = +15.00

joint 10



joint 43



Rod 52

joint 43 → 27

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +75.00 kN
at 43: Fx = +0.00 Fy = +75.00
at 27: Fx = +0.00 Fy = -75.00

joint 43



joint 27



Rod 53

joint 11 → 44

compression

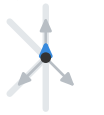


L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -45.00 kN
at 11: Fx = +0.00 Fy = -45.00
at 44: Fx = +0.00 Fy = +45.00

joint 11



joint 44



Rod 54

joint 44 → 28

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +105.00 kN
at 44: Fx = +0.00 Fy = +105.00
at 28: Fx = +0.00 Fy = -105.00

joint 44



joint 28



Rod 55

joint 12 → 45

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -75.00 kN
at 12: Fx = +0.00 Fy = -75.00
at 45: Fx = +0.00 Fy = +75.00

joint 12



joint 45



Rod 56

joint 45 → 29

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +135.00 kN
at 45: Fx = +0.00 Fy = +135.00
at 29: Fx = +0.00 Fy = -135.00

joint 45



joint 29



Rod 57

joint 13 → 46

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -105.00 kN
at 13: Fx = +0.00 Fy = -105.00
at 46: Fx = +0.00 Fy = +105.00

joint 13



joint 46



Rod 58

joint 46 → 30

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +165.00 kN
at 46: Fx = +0.00 Fy = +165.00
at 30: Fx = +0.00 Fy = -165.00

joint 46



joint 30



Rod 59

joint 14 → 47

compression

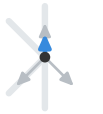


L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -135.00 kN
at 14: Fx = +0.00 Fy = -135.00
at 47: Fx = +0.00 Fy = +135.00

joint 14



joint 47



Rod 60

joint 47 → 31

tension



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +195.00 kN
at 47: Fx = +0.00 Fy = +195.00
at 31: Fx = +0.00 Fy = -195.00

joint 47



joint 31



Rod 61

joint 15 → 48

compression

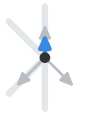


L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -165.00 kN
at 15: Fx = +0.00 Fy = -165.00
at 48: Fx = +0.00 Fy = +165.00

joint 15



joint 48



Rod 62

joint 48 → 32

tension

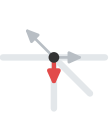


L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +225.00 kN
at 48: Fx = +0.00 Fy = +225.00
at 32: Fx = +0.00 Fy = -225.00

joint 48



joint 32



Rod 63

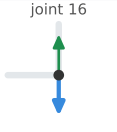
joint 16 → 49

compression



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = -450.00 kN
at 16: Fx = +0.00 Fy = -450.00
at 49: Fx = +0.00 Fy = +450.00

joint 16



joint 49



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

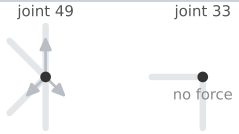
Rod 64

joint 49 → 33

no force



L = 4.000 m θ = 90.0°
cos θ = +0.000 sin θ = +1.000
N = +0.00 kN
at 49: Fx = +0.00 Fy = +0.00
at 33: Fx = +0.00 Fy = +0.00



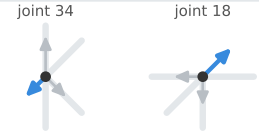
Rod 65

joint 34 → 18

compression



L = 5.657 m θ = 45.0°
cos θ = +0.707 sin θ = +0.707
N = -318.20 kN
at 34: Fx = -225.00 Fy = -225.00
at 18: Fx = +225.00 Fy = +225.00



Rod 66

joint 34 → 1

tension



L = 5.657 m θ = 315.0°
cos θ = +0.707 sin θ = -0.707
N = +318.20 kN
at 34: Fx = +225.00 Fy = -225.00
at 1: Fx = -225.00 Fy = +225.00



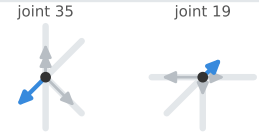
Rod 67

joint 35 → 19

compression



L = 5.657 m θ = 45.0°
cos θ = +0.707 sin θ = +0.707
N = -275.77 kN
at 35: Fx = -195.00 Fy = -195.00
at 19: Fx = +195.00 Fy = +195.00



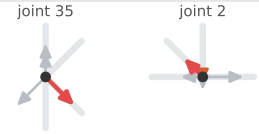
Rod 68

joint 35 → 2

tension



L = 5.657 m θ = 315.0°
cos θ = +0.707 sin θ = -0.707
N = +275.77 kN
at 35: Fx = +195.00 Fy = -195.00
at 2: Fx = -195.00 Fy = +195.00



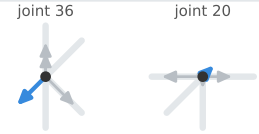
Rod 69

joint 36 → 20

compression



L = 5.657 m θ = 45.0°
cos θ = +0.707 sin θ = +0.707
N = -233.35 kN
at 36: Fx = -165.00 Fy = -165.00
at 20: Fx = +165.00 Fy = +165.00



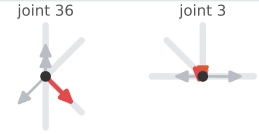
Rod 70

joint 36 → 3

tension



L = 5.657 m θ = 315.0°
cos θ = +0.707 sin θ = -0.707
N = +233.35 kN
at 36: Fx = +165.00 Fy = -165.00
at 3: Fx = -165.00 Fy = +165.00



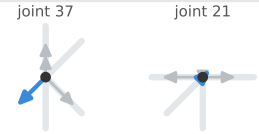
Rod 71

joint 37 → 21

compression



L = 5.657 m θ = 45.0°
cos θ = +0.707 sin θ = +0.707
N = -190.92 kN
at 37: Fx = -135.00 Fy = -135.00
at 21: Fx = +135.00 Fy = +135.00



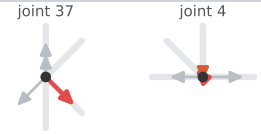
Rod 72

joint 37 → 4

tension



L = 5.657 m θ = 315.0°
cos θ = +0.707 sin θ = -0.707
N = +190.92 kN
at 37: Fx = +135.00 Fy = -135.00
at 4: Fx = -135.00 Fy = +135.00



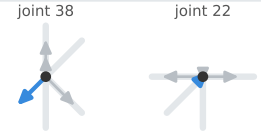
Rod 73

joint 38 → 22

compression



L = 5.657 m θ = 45.0°
cos θ = +0.707 sin θ = +0.707
N = -148.49 kN
at 38: Fx = -105.00 Fy = -105.00
at 22: Fx = +105.00 Fy = +105.00



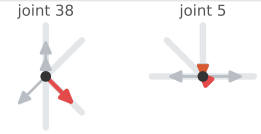
Rod 74

joint 38 → 5

tension



L = 5.657 m θ = 315.0°
cos θ = +0.707 sin θ = -0.707
N = +148.49 kN
at 38: Fx = +105.00 Fy = -105.00
at 5: Fx = -105.00 Fy = +105.00



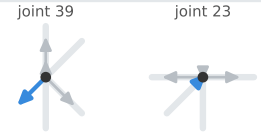
Rod 75

joint 39 → 23

compression



L = 5.657 m θ = 45.0°
cos θ = +0.707 sin θ = +0.707
N = -106.07 kN
at 39: Fx = -75.00 Fy = -75.00
at 23: Fx = +75.00 Fy = +75.00



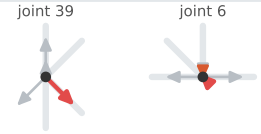
Rod 76

joint 39 → 6

tension



L = 5.657 m θ = 315.0°
cos θ = +0.707 sin θ = -0.707
N = +106.07 kN
at 39: Fx = +75.00 Fy = -75.00
at 6: Fx = -75.00 Fy = +75.00



Rod 77

joint 40 → 24

compression



L = 5.657 m θ = 45.0°
cos θ = +0.707 sin θ = +0.707
N = -63.64 kN
at 40: Fx = -45.00 Fy = -45.00
at 24: Fx = +45.00 Fy = +45.00



Rod 78

joint 40 → 7

tension



L = 5.657 m θ = 315.0°
cos θ = +0.707 sin θ = -0.707
N = +63.64 kN
at 40: Fx = +45.00 Fy = -45.00
at 7: Fx = -45.00 Fy = +45.00



Rod 79

joint 41 → 25

compression



L = 5.657 m θ = 45.0°
cos θ = +0.707 sin θ = +0.707
N = -21.21 kN
at 41: Fx = -15.00 Fy = -15.00
at 25: Fx = +15.00 Fy = +15.00



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

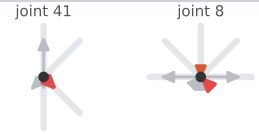
Rod 80

joint 41 → 8

tension



L = 5.657 m θ = 315.0°
cos θ = +0.707 sin θ = -0.707
N = +21.21 kN
at 41: Fx = +15.00 Fy = -15.00
at 8: Fx = -15.00 Fy = +15.00



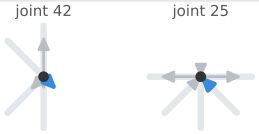
Rod 81

joint 42 → 25

compression



L = 5.657 m θ = 135.0°
cos θ = -0.707 sin θ = +0.707
N = -21.21 kN
at 42: Fx = +15.00 Fy = -15.00
at 25: Fx = -15.00 Fy = +15.00



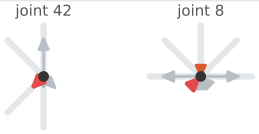
Rod 82

joint 42 → 8

tension



L = 5.657 m θ = 225.0°
cos θ = -0.707 sin θ = -0.707
N = +21.21 kN
at 42: Fx = -15.00 Fy = -15.00
at 8: Fx = +15.00 Fy = +15.00



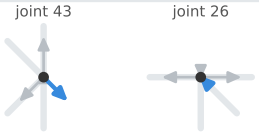
Rod 83

joint 43 → 26

compression



L = 5.657 m θ = 135.0°
cos θ = -0.707 sin θ = +0.707
N = -63.64 kN
at 43: Fx = +45.00 Fy = -45.00
at 26: Fx = -45.00 Fy = +45.00



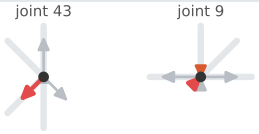
Rod 84

joint 43 → 9

tension



L = 5.657 m θ = 225.0°
cos θ = -0.707 sin θ = -0.707
N = +63.64 kN
at 43: Fx = -45.00 Fy = -45.00
at 9: Fx = +45.00 Fy = +45.00



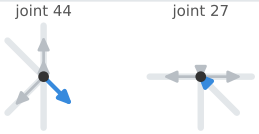
Rod 85

joint 44 → 27

compression



L = 5.657 m θ = 135.0°
cos θ = -0.707 sin θ = +0.707
N = -106.07 kN
at 44: Fx = +75.00 Fy = -75.00
at 27: Fx = -75.00 Fy = +75.00



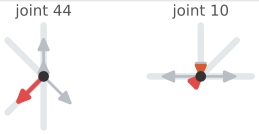
Rod 86

joint 44 → 10

tension



L = 5.657 m θ = 225.0°
cos θ = -0.707 sin θ = -0.707
N = +106.07 kN
at 44: Fx = -75.00 Fy = -75.00
at 10: Fx = +75.00 Fy = +75.00



Rod 87

joint 45 → 28

compression



L = 5.657 m θ = 135.0°
cos θ = -0.707 sin θ = +0.707
N = -148.49 kN
at 45: Fx = +105.00 Fy = -105.00
at 28: Fx = -105.00 Fy = +105.00



Rod 88

joint 45 → 11

tension



L = 5.657 m θ = 225.0°
cos θ = -0.707 sin θ = -0.707
N = +148.49 kN
at 45: Fx = -105.00 Fy = -105.00
at 11: Fx = +105.00 Fy = +105.00



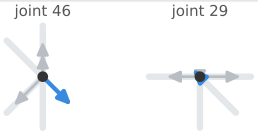
Rod 89

joint 46 → 29

compression



L = 5.657 m θ = 135.0°
cos θ = -0.707 sin θ = +0.707
N = -190.92 kN
at 46: Fx = +135.00 Fy = -135.00
at 29: Fx = -135.00 Fy = +135.00



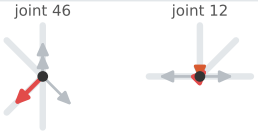
Rod 90

joint 46 → 12

tension



L = 5.657 m θ = 225.0°
cos θ = -0.707 sin θ = -0.707
N = +190.92 kN
at 46: Fx = -135.00 Fy = -135.00
at 12: Fx = +135.00 Fy = +135.00



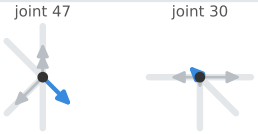
Rod 91

joint 47 → 30

compression



L = 5.657 m θ = 135.0°
cos θ = -0.707 sin θ = +0.707
N = -233.35 kN
at 47: Fx = +165.00 Fy = -165.00
at 30: Fx = -165.00 Fy = +165.00



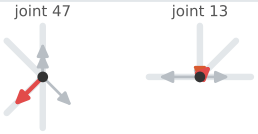
Rod 92

joint 47 → 13

tension



L = 5.657 m θ = 225.0°
cos θ = -0.707 sin θ = -0.707
N = +233.35 kN
at 47: Fx = -165.00 Fy = -165.00
at 13: Fx = +165.00 Fy = +165.00



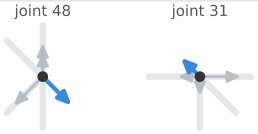
Rod 93

joint 48 → 31

compression



L = 5.657 m θ = 135.0°
cos θ = -0.707 sin θ = +0.707
N = -275.77 kN
at 48: Fx = +195.00 Fy = -195.00
at 31: Fx = -195.00 Fy = +195.00



Rod 94

joint 48 → 14

tension



L = 5.657 m θ = 225.0°
cos θ = -0.707 sin θ = -0.707
N = +275.77 kN
at 48: Fx = -195.00 Fy = -195.00
at 14: Fx = +195.00 Fy = +195.00



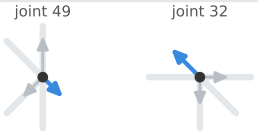
Rod 95

joint 49 → 32

compression



L = 5.657 m θ = 135.0°
cos θ = -0.707 sin θ = +0.707
N = -318.20 kN
at 49: Fx = +225.00 Fy = -225.00
at 32: Fx = -225.00 Fy = +225.00



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 96

joint 49 → 15

tension



```
L = 5.657 m     θ = 225.0°
cos θ = -0.707   sin θ = -0.707
N = +318.20 kN
at 49:  Fx = -225.00   Fy = -225.00
at 15:  Fx = +225.00   Fy = +225.00
```

joint 49



joint 15





Rod 8

utilisation 0.73

normal stress (H.3.4)

```
Rod 8
=====
Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 945,000 N / 6,156 mm²
      = 153.51 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.51 / 211.50
      = 0.726

 4. Governs
      the largest ratio
      = normal stress (H.3.4)
      = 0.726

utilisation 0.726 -- OK
```



Rod 7

utilisation 0.73

normal stress (H.3.4)

```
Rod 7
=====
Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 945,000 N / 6,156 mm²
      = 153.51 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.51 / 211.50
      = 0.726

 4. Governs
      the largest ratio
      = normal stress (H.3.4)
      = 0.726

utilisation 0.726 -- OK
```